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Final Report for Computational Astrophysics

Course of 3rd semester - Physics of Data

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ABSTRACT

Context. The study of exoplanets combines data science, atmospheric modeling, and observational techniques to detect and characterize planetary systems.
Aims. This project explores automated exoplanet detection using machine learning on Kepler light curves and atmospheric characterization through Bayesian retrieval of JWST transmission spectra.
Methods. Support Vector Machines and Neural Networks were applied to classify transit signals in Kepler data. TauREx-3 was used for atmospheric retrieval of MASCARA-3b synthetic spectra and WASP-39b JWST observations.
Results. Machine learning achieved 99% precision on clean data but dropped to 56% on augmented datasets, highlighting detection challenges. Atmospheric retrieval successfully recovered planetary parameters for synthetic data and identified H_2O (6.3%), CO_2 (0.14%), and radius (1.22 R_J) for WASP-39 b, though simplified isothermal models showed systematic temperature offsets.
Conclusions. These results demonstrate both the potential and limitations of machine learning for transit vetting and the importance of iterative refinement in atmospheric retrieval when model assumptions are inadequate.

Key words. exoplanets, atmosphere, retrieval, spectrum, svm, nn, batman, taurex, transit, composition

1. Introduction

The discovery of exoplanets has transformed astrophysics over the past three decades, driven by ground-based surveys and space missions such as Kepler, TESS, and JWST. Hot Jupiters and hot Saturns are particularly favorable observational targets due to their large radii, inflated atmospheres, and short orbital periods, which produce deep, frequent transits. Bright host stars further enhance detectability by enabling high signal-to-noise photometry and spectroscopy.

This study focuses on WASP-39b, a highly inflated hot Saturn discovered by the Wide Angle Search for Planets (WASP) survey in 2011 (Faedi et al. 2011). WASP-39b orbits a G8-type star every 4.05 days and possesses one of the lowest known densities for a gas giant exoplanet, making it an exceptional target for atmospheric characterization. Early ground-based and HST observations detected sodium absorption (Nikolov et al. 2016), establishing WASP-39b as a promising candidate for detailed atmospheric study. The advent of JWST has revolutionized our understanding of this system: observations across multiple instruments (NIRISS, NIRCam, NIRSpec G395H and PRISM, MIRI) spanning 0.5–5.2 μm revealed absorption features from H_2O, CO_2, CO, SO_2, Na, and K (Alderson et al. 2023; Rustamkulov et al. 2023; Tsai et al. 2023; Carter et al. 2024). The detection of SO_2 was particularly significant as it provided the first direct evidence for photochemically-driven disequilibrium chemistry in an exoplanet atmosphere (Tsai et al. 2023).

Despite this wealth of observational data, significant uncertainties remain in our understanding of WASP-39b's atmospheric composition and structure. Retrieval analyses using different radiative transfer codes, cloud parameterizations, and chemical assumptions yield substantially different inferred metallicities, C/O ratios, and even molecular detections (Lueber et al. 2024; Ma et al. 2025). The interpretation of the 4.1 μm spectral feature remains particularly debatable, with some studies attributing it to photochemically produced SO_2 (Tsai et al. 2023) while others suggest SiO gas and silicate clouds can reproduce the same features within an equilibrium chemistry framework (Ma et al. 2025). These model-dependent discrepancies highlight a fundamental challenge in exoplanet atmospheric science: strong degeneracies between cloud properties and molecular abundances can bias retrieved atmospheric parameters by orders of magnitude, making it difficult to extract robust physical insights from even the highest-quality spectra.

This work addresses two complementary aspects of exoplanet characterization. First, we develop and validate machine learning methods for automated transit detection and candidate vetting using Kepler light curves. MASCARA-3b (also known as KELT-24b) (Hjorth et al. 2019) serves as a benchmark for understanding realistic transit signals in our validation process. MASCARA-3b is a hot Jupiter discovered by the MASCARA project (Multi-site All-Sky CAmeRA) (Snellen et al. 2013) and independently by KELT (Kilodegree Extremely Little Telescope) via the transit method (Talens, G. J. J. et al. 2018) orbiting a bright F-type star with a period of 5.55 days. The combination of a bright host star and large planetary radius makes MASCARA-3b well-suited for understanding transit detection challenges.

Second, we perform atmospheric retrieval on WASP-39b's JWST transmission spectrum to critically assess the reliability of retrieved parameters and explore the impact of different modeling assumptions. This study evaluates both the capabilities and constraints of current approaches, providing a grounded frame-

work for interpreting atmospheric results and pinpointing areas for model refinement.

The structure of this report is as follows:

Section 2: summarizes the known stellar and planetary parameters for MASCARA-3b and WASP-39b from the literature and discusses how these characteristics influence detectability. **Section 3:** presents the machine learning classification framework for transit detection, comparing Support Vector Machine and Artificial Neural Network approaches validated on Kepler light curves. **Section 4:** reviews the atmospheric literature on WASP-39b, describes our atmospheric retrieval methodology using TauREx-3 and interpretation of results. **Section 5:** concludes by integrating insights from both the ML detection pipeline and atmospheric characterization, highlighting key scientific results, proposing methodological improvements, and discussing implications for upcoming missions.

2. Planetary System Parameters

Planetary and stellar parameters were obtained from the Encyclopaedia of Exoplanetary Systems and cross-checked against primary literature. WASP-39b was selected for atmospheric analysis due to availability of JWST transmission spectra and published atmospheric studies, fulfilling Assignment 3 requirements.

2.1. MASCARA-3 System

MASCARA-3b is a 5.18 M_J hot Jupiter orbiting an F7 IV–V star with an orbital period of 5.551 days. The host star's brightness (V = 8.4) and effective temperature (6347 K) enable high-precision photometry despite the moderate planet-to-star flux ratio. The large planetary radius (1.27 R_J) and high equilibrium temperature (1458 K) produce a substantial atmospheric scale height, enhancing transit depth and spectral feature amplitudes.

The bright host star enables ground-based detection via the MASCARA wide-field survey despite the relatively small transit depth (~0.8%). The large planetary radius and high temperature create an extended atmosphere with scale height $H \approx 700$ km, enhancing atmospheric transmission signals at wavelengths corresponding to molecular absorption bands. Here, we can see the hypothetical visualization of the MASCARA-3 system by NASA.

2.2. WASP-39 System

WASP-39b was the first exoplanet that was studied by JWST in 2022! It is a 0.28 M_J inflated hot Saturn orbiting a G8-type star every 4.055 days. The planet's extremely low density (0.14 g cm^{-3}) yields one of the largest atmospheric scale heights among known transiting exoplanets, enabling detailed transmission spectroscopy. The moderately bright host star (V = 12.1) and favorable planet-star radius ratio ($R_p/R_\star \approx 0.14$) produce deep transits (~2.5%) with strong atmospheric signals observable from space.

The extremely low density is due to the fact that its atmosphere is very extended so its transits produce strong signals, and its star does not add much noise. So the atmospheric scale hight $H \approx 1600$ km is among the largest for any known transiting planet. Combined with deep transits and the relatively quiet G-type host, this produces exceptionally strong atmospheric features detectable across optical and infrared wavelengths. The system also has southern declination (−3°) which means that it provided excellent visibility for both HST and JWST, making WASP-39b a benchmark atmospheric target. Here, we can see the hypothetical visualization of the WASP-39 system by NASA, as well.

These characteristics make MASCARA-3b ideal for testing automated detection algorithms on well-characterized transits, while WASP-39b serves as a validation target for atmospheric retrieval methods as we can find high-quility spectra from JWST.

Planet: MASCARA-3b / KELT-24b

Parameter	Value
Mass	5.18 ± 0.22 M_J
Semi-major axis α	0.0700.009 AU
Orbital Period	5.551 ± 0.000002 d
e	0.085 ± 0.023
T_{peri} (JD)	2457146.6 ± 0.7
Radius	1.272 ± 0.022 R_J
T_{eq}	1458 ± 16 K
i	$89.16^{+0.6}_{-0.77}$
Discovery	Transit (2019)

Star: HD 93148

Parameter	Value
Distance	96.79 ± 0.03 pc
Spectral Type	F7 IV–V
Apparent Magnitude V	8.4
Mass	1.31 ± 0.05 $M_\odot$
Age	$2.81^{+4.0}_{-0.8}$ Gyr
T_{eff}	6347 ± 60 K
Radius	1.27 ± 0.02 $R_\odot$
Metallicity [Fe/H]	$0.17^{+0.07}_{-0.06}$

Table 1: Basic Parameters of the MASCARA-3 System

3. Detection of Transit Light Curves Using Machine Learning

Transit photometry surveys like Kepler and TESS generate enormous catalogs requiring automated vetting to identify genuine planetary candidates among false positives from eclipsing binaries, stellar variability, and instrumental artifacts. Machine learning offers scalable pattern recognition capable of learning subtle transit signatures directly from light curve data.

We applied supervised classification to Kepler light curves containing 3197 flux measurements per star. The dataset exhibits severe class imbalance: the training set contains 5087 light curves with only 37 confirmed planets (0.7%), while the validation set has 570 light curves with 5 planets. Two versions were analyzed: `data_no_injection` containing authentic Kepler detections, and `data_injected` with synthetic transits added to simulate weaker signals near the detection threshold.

Preprocessing Pipeline: Raw light curves were transformed through four sequential operations to extract transit signatures while suppressing noise:

Planet: WASP-39b

Parameter	Value
Mass	$0.28 \pm 0.03\ M_J$
Semi-major axis α	0.0486 ± 0.0005 AU
Orbital Period	4.0553 ± 0.000009 d
e	0.0 (circular)
T_{peri} (JD)	2457146.6 ± 0.7
Radius	$1.27 \pm 0.04\ R_J$
T_{eq}	1120 K
i	87.83 ± 0.25
Discovery	Transit (2011)

Star: WASP-39

Parameter	Value
Distance	230 ± 80 pc
Spectral Type	G8
Apparent Magnitude V	12.1
Mass	$0.93 \pm 0.03\ M_\odot$
Age	8.5 - 9 Gyr
T_{eff}	5400 ± 150 K
Radius	$0.895 \pm 0.023\ R_\odot$
Metallicity [Fe/H]	-0.12 ± 0.1

Table 2: Basic Parameters of the WASP-39 System

1. **Fourier transformation:** Time-domain flux was converted to frequency space via Fast Fourier Transform. Since planetary transits are periodic events occurring at fixed orbital intervals, the Fourier transform decomposes the time-series light curve into constituent frequencies, where periodic signals appear as distinct peaks. By converting the light curve into frequencies, the regular transit signal becomes stronger and easier to spot, while random noise is spread thin and less noticeable. The symmetry of the Fourier transform for real-valued inputs allows retention of only the first half of the spectrum 1598 features from original 3197), reducing dimensionality without information loss.
2. **Normalization:** Each Fourier spectrum was scaled to unit L2 norm, removing dependence on stellar brightness differences and ensuring uniform feature magnitudes across the dataset.
3. **Gaussian smoothing:** Convolution with a Gaussian kernel suppressed high-frequency noise while preserving transit features spanning 20–40 data points. The kernel width $\sigma = 10$ represents a commonly used empirically optimized balance: smaller values ($\sigma < 5$) retain excessive noise, while larger values ($\sigma > 20$) blur genuine transit signals.
4. **Standardization:** Features were centered to zero mean and scaled to unit variance, accelerating gradient-based optimization by preventing large-magnitude features from dominating the loss landscape.

This pipeline reduces raw 3197-element flux vectors to 1598-element feature vectors optimized for machine learning classification.

3.1. Methodology

Planetary transits produce characteristic periodic flux decreases with specific ingress/egress profiles determined by planetary radius, orbital inclination, and limb darkening. However, distinguishing these from astrophysical mimics (eclipsing binaries, stellar spots) and systematic effects requires recognizing complex, high-dimensional patterns. Support Vector Machines and Neural Networks excel at learning nonlinear decision boundaries in such feature spaces, making them well-suited for automated transit classification.

Class Imbalance Challenge: The extreme rarity of confirmed planets (0.7% positive examples) biases models toward predicting the majority class. Without mitigation, classifiers achieve high accuracy by labeling all examples as non-planets while completely failing to detect actual transits. We address this through algorithm-specific strategies: SVMs were evaluated with default parameters to assess baseline performance, while Neural Networks employed SMOTE (Synthetic Minority Oversampling Technique) to balance training data by generating synthetic positive examples via interpolation.

False Positive Sources: Grazing eclipsing binaries, stellar activity (flares, rotation), and centroid motion from blended background eclipsing binaries can mimic transit light curves. Distinguishing these requires either multi-wavelength observations, high-resolution imaging, or spectroscopic follow-up. Machine learning accelerates initial vetting by identifying promising candidates for expensive confirmatory observations, but cannot definitively confirm planetary nature from photometry alone.

3.1.1. Support Vector Machine Configuration

SVMs construct optimal separating hyperplanes by maximizing the margin between classes, with support vectors defining the decision boundary. Three kernel functions were tested to assess sensitivity to feature space geometry:

- **Linear kernel:** Assumes linear separability in the 1598-dimensional feature space.
- **Radial Basis Function (RBF):** Maps features to infinite-dimensional space, enabling arbitrarily complex nonlinear boundaries. The parameter γ controls kernel width.
- **Polynomial (degree 4):** Provides intermediate flexibility between linear and RBF kernels.

All models used `scikit-learn` defaults: $C = 1.0$ (regularization), $\gamma = 1/(n_{\text{features}} \times \text{Var}(\mathbf{X}))$. No class weighting was applied, allowing assessment of raw performance under severe imbalance.

Evaluation Metrics: Precision (True Positives / [True Positives + False Positives]) was prioritized over accuracy due to class imbalance.

Results:

Not injected data: All kernels achieved very high precision (>99%), with RBF and polynomial eliminating false positives entirely. The strong performance indicates Fourier-based features effectively capture periodic transit signatures when signal-to-noise is high.

Injected data: Precision collapsed to ~56% across all kernels. RBF and polynomial exhibited extreme behavior (predicting nearly all examples as positive), achieving high recall (99.6–99.8%) at the cost of massive false positive rates. This demonstrates kernel choice matters less than the fundamental difficulty

Kernel	Dataset	Precision	Confusion Matrix (Dev)
Linear	not injected	99.10%	[[552, 13], [5, 0]]
RBF	not injected	99.12%	[[565, 0], [5, 0]]
Poly (d=4)	not injected	99.12%	[[565, 0], [5, 0]]
Linear	injected	55.86%	[[201, 245], [262, 310]]
RBF	njected	56.32%	[[4, 442], [2, 570]]
Poly (d=4)	njected	56.20%	[[1, 445], [1, 571]]

Table 3: SVM - Confusion matrix [[TN, FP], [FN, TP]]

of detecting weak signals. With ~56% being the highest, it represents the information limit of Fourier-derived features without physical transit parameters (depth, duration, shape).

3.1.2. Neural Network Architecture

A fully connected feedforward network was implemented in TensorFlow:

- **Input layer:** 1598 features (post-preprocessing)
- **Hidden layer 1:** 128 neurons, ReLU activation
- **Dropout (rate = 0.3):** Randomly deactivates 30% of neurons during training
- **Hidden layer 2:** 64 neurons, ReLU activation
- **Dropout (rate = 0.3)**
- **Output layer:** 1 neuron, sigmoid activation (binary classification)

Training Configuration: Adam optimizer (adaptive learning rates), binary cross-entropy loss, 50 epochs, batch size 32. SMOTE oversampling balanced the training set by generating synthetic minority-class examples via k-nearest neighbor interpolation.

Justification: ReLU activations prevent vanishing gradients, allowing effective training of deep networks. Dropout improves generalization by reducing overfitting. The network gradually reduces dimensionality (1598 → 128 → 64 → 1) to learn increasingly compact feature representations.

Results:

Dataset	Split	Precision	Confusion Matrix
not injected	Train	97.37%	[[5049, 1], [0, 37]]
not injected	Dev	41.67%	[[558, 7], [0, 5]]
injected	Train	57.89%	[[265, 1961], [165, 2696]]
injected	Dev	55.44%	[[0, 254], [0, 316]]

Table 4: NN - Confusion matrix [[TN, FP], [FN, TP]]

Not injected data: Severe overfitting (97.4% train → 41.7% dev). The network memorized training examples rather than learning generalizable transit features. Despite dropout, the small dataset size (37 positive examples after SMOTE augmentation) proved insufficient for robust neural network training.

Injected data: Train and dev precision converged to ~56%, indicating no overfitting. However, overall performance remained poor, matching the SVM ceiling. The network learned to predict nearly all examples as positive (dev confusion matrix: 0 true negatives), achieving perfect recall but minimal precision.

3.2. Results and Comparison

Performance Summary:

- **Not injected data:** SVM dominates (99% vs. 42% dev precision). Small sample sizes favor simpler models less prone to overfitting.
- **Injected data:** Methods converge to identical performance (~56%). The problem difficulty, not algorithm choice, determines the ceiling.

1. **Feature limitations:** Fourier coefficients capture periodicity but discard transit shape information (ingress/egress profiles, depth variations). Physical feature extraction (transit depth, duration, impact parameter) would likely improve performance on weak signals.
2. **Class imbalance mitigation essential:** Without SMOTE or class weighting, models predict only the majority class. Future work should explore cost-sensitive learning or anomaly detection frameworks.
3. **Overfitting vs. underfitting:** Neural networks require more data than available. SVMs show greater robustness to small sample sizes, making them preferable for survey-scale vetting pipelines operating on limited confirmed examples.

Improvements: Future work could focus on using physically motivated transit parameters as model inputs, rather than relying only on Fourier features. Model performance could also be improved by combining SVM and neural network predictions in an ensemble approach, or by exploring One-Class SVMs trained only on confirmed transit signals (Roche 2024). Finally, convolutional neural networks designed for time-series data could be tested to better capture the characteristic shape of transit events.

4. Atmospheric Study on WASP-39b

4.1. Observed Atmospheric Findings

WASP-39b's atmospheric characterization represents a landmark achievement in exoplanet science, mentioned in (Tsai et al. 2023; Rustamkulov et al. 2023; Alderson et al. 2023). High-precision transmission spectra from JWST (NIRISS, NIRCam, NIRSpec G395H, NIRSpec PRISM, and MIRI) spanning 0.5–5.2 μm reveal prominent absorption features from multiple molecular species (Roy-Perez et al. 2025; Lueber et al. 2024; Ma et al. 2025).

The most robust detections across all JWST instruments are H_2O and CO_2 (Tsai et al. 2023; Alderson et al. 2023), with additional features clearly attributed to Na, K, and at least one sulfur- or silicon-bearing species (Lueber et al. 2024; Ma et al. 2025).

These observations consistently indicate a hydrogen-dominated atmosphere with mean molecular weight $\mu \approx 2.3$, near-solar to super-solar metallicity, and characteristic terminator temperatures of 1000–1200 K (Gkouvelis et al. 2026; Ma et al. 2025). The muted amplitude of spectral features requires the presence of clouds or hazes, with JWST's broad wavelength coverage enabling constraints on the wavelength-dependence of aerosol extinction that were impossible with earlier HST/Spitzer observations (Roy-Perez et al. 2025; Lueber et al. 2024). Earlier HST and Spitzer-era models often approximated clouds as grey or Rayleigh-like opacity (Nikolov et al. 2016), limited by their narrow wavelength ranges, whereas JWST's coverage from 0.5–12 μm allows for detailed aerosol characterization (Roy-Perez et al. 2025).

A key area of active debate concerns the interpretation of features in the 4–5 μm region. Some analyses identify SO_2 absorption at 4.1 μm, interpreting this as evidence for UV-driven sulfur photochemistry—the first such direct detection in an exoplanet atmosphere (Gkouvelis et al. 2026), first detected in (Tsai et al. 2023).

However, alternative scenarios suggest that SiO gas and silicate clouds ($MgSiO_3$, SiO_2) can reproduce not only the 4.1 μm feature but also mid-infrared continuum structure at 7.5 and 8.8 μm, potentially indicating an atmosphere closer to chemical equilibrium when silicon-bearing species are included (Ma et al. 2025). This ambiguity highlights the sensitivity of molecular identifications and derived abundance ratios (C/O, S/O) to assumptions about cloud opacity and chemical priors.

Beyond composition, some JWST modes suggest non-isothermal transit chords are necessary to fit the data, indicating vertical temperature gradients at the terminator (Lueber et al. 2024). JWST's unprecedented wavelength coverage and sensitivity establish WASP-39b as a benchmark hot-Saturn, though detailed questions about sulfur versus silicon chemistry and cloud microphysics remain under active investigation.

4.2. Models Used in the Literature

Published analyses of WASP-39b employ a variety of modeling approaches that can be broadly categorized by their treatment of atmospheric physics, chemistry, and retrieval strategy. This section reviews the main frameworks used to interpret JWST observations, along with their underlying assumptions and limitations.

4.2.1. Forward Modeling and Radiative Transfer

Most studies adopt 1D plane-parallel transmission models assuming hydrostatic equilibrium, a parameterized temperature–pressure (T–p) profile, and a vertically mixed set of absorbers. Radiative transfer is typically solved using either line-by-line or correlated-k methods.

One widely used tool is the Planetary Spectrum Generator (PSG), whose radiative-transfer engine PUMAS computes transmission spectra in a 1D plane-parallel geometry (Roy-Perez et al. 2025). PUMAS includes opacities for key molecules (H_2O, CO_2, CO, SO_2, etc.) from high-temperature line lists, together with continuum processes such as Rayleigh scattering, collision-induced absorption by H_2–H_2 and H_2–He, refraction, UV broad absorption, and multiple scattering. Bayesian retrievals using PSG with nested sampling (MultiNest) can simultaneously constrain gas abundances and aerosol properties, with priors on planetary parameters ($R_\star$, D_{pl}, log g) typically taken from literature values such as Faedi et al. (2011).

Other works employ TauREx-3 (Al-Refaie et al. 2021), frequently combined with chemical-equilibrium or photochemical models to define prior abundances or training grids (Ma et al. 2025; Lueber et al. 2024).

Some studies pre-compute transmission spectra on grids spanning metallicity, C/O, and cloud parameters using codes like petitRADTrans (Mollière et al. 2019), then interpolate via machine-learning regressors for fast approximate Bayesian inference (Lueber et al. 2024).

These frameworks generally assume constant gravity over all atmospheric layers and a parametric terminator T–p profile that can be isothermal or allow for mild vertical structure.

Semi-analytical approaches offer additional insight by prescribing the pressure-dependence of opacity $\kappa(P, \lambda)$ analytically. For example, (Gkouvelis et al. 2026) developed a non-isobaric transmission formula and demonstrated significantly improved fits to WASP-39b compared to isobaric formulations. While computationally efficient and interpretable, such models do not include full chemistry or 3D atmospheric structure.

4.2.2. Chemical Modeling Strategies

Two main approaches to atmospheric chemistry appear in the literature:

Free retrievals treat the volume mixing ratios of selected species (H_2O, CO_2, CO, SO_2/SiO, Na, K, CH_4, H_2S, etc.) as independent parameters with wide log-uniform priors (Lueber et al. 2024). This approach maximizes flexibility but can lead to physically implausible abundance combinations and does not guarantee chemical self-consistency.

Chemically informed retrievals use equilibrium chemistry models to identify plausible species and constrain priors before performing retrieval. (Ma et al. 2025) introduced a hybrid strategy that iteratively combines equilibrium chemistry calculations, cloud microphysics, and free retrievals with TauREx, cycling through these steps until the derived atmospheric state converges to a self-consistent solution (Ma et al. 2025).

This approach emphasizes physical plausibility by constraining priors using equilibrium chemistry and cloud microphysics before the retrieval, but remains sensitive to assumptions about whether the atmosphere is in equilibrium or has strong disequilibrium chemistry.

A specific example is the analysis by Carter et al., who employed a 1D radiative-convective photochemical-equilibrium model at 10× solar metallicity and sub-solar C/O = 0.35, then post-processed with parameterized clouds and hazes to match JWST spectra. Their retrieval framework used nested sampling (MultiNest) to explore posterior distributions for approximately 15 free parameters including molecular abundances, cloud properties, and atmospheric temperature structure. This approach assumes global averaging and that the atmosphere has reached chemical and photochemical equilibrium, but it does not fully couple aerosol formation with atmospheric dynamics.

4.2.3. Cloud and Aerosol Parameterizations

Cloud modeling is critical for WASP-39b, as JWST data unambiguously require aerosols to explain muted spectral features. Clouds are typically parameterized as either grey (wavelength-independent opacity) or non-grey (wavelength-dependent), with some models specifying particle size distributions, compositions (e.g., silicates, sulfides), and vertical profiles (Lueber et al. 2024; Ma et al. 2025).

JWST observations favor aerosols with varying opacity consistent with micron-sized Mie scattering, and Bayesian model comparison strongly prefers cloudy models over cloud-free ones across all instruments. However, the choice of cloud parameterization drastically affects retrieval results. Non-grey cloud opacity can mimic molecular absorption features—particularly water vapor—leading to false detections and biasing abundance estimates by orders of magnitude (Lueber et al. 2024). So, since it's harder to uniquely determine both cloud properties and gas composition because they can mimic each other in the spectrum, metallicity and C/O estimates become highly uncertain.

4.2.4. Statistical Framework and Data Reduction

Most WASP-39b studies use Bayesian nested sampling to explore high-dimensional parameter spaces and compute model evidences for comparison (Roy-Perez et al. 2025; Ma et al. 2025; Carter et al. 2024). Nested sampling with MultiNest or similar algorithms provides robust marginalized posteriors and handles multi-modal distributions that can arise from parameter degeneracies.

Recent work has also highlighted the impact of data-reduction choices on retrieval outcomes. (Schleich et al. 2024) analyzed how different pipeline reductions of the same JWST NIRSpec PRISM transit affect posteriors, classifying parameters as stable (H_2O, CO_2), weakly constrained (CO, CH_4), or unstable (SO_2, C_2H_2) depending on the input spectrum.

They demonstrate that posterior shapes for some species like SO_2 and C_2H_2 can become heavy-tailed and exhibit substantial parameter shifts across different data reductions. This demonstrates that pipeline-level systematics are now a limiting factor for detecting minor species and determining precise abundances.

4.2.5. Key Assumptions and Limitations

While these models have enabled transformative insights into WASP-39b's atmosphere, several important assumptions and limitations should be recognized:

- **1D geometry:** All published models assume a 1D limb-average atmosphere, neglecting day-night thermal contrasts and morning-evening terminator asymmetries that could lead to longitudinal composition gradients. For example, Carter et al. explicitly note that their 1D limb-average geometry neglects day-night and morning-evening terminator variations that could be significant in hot Saturns (Carter et al. 2024).
- **Parameterized clouds:** Most retrievals use parameterized rather than self-consistent cloud models. Clouds are prescribed with simple opacity laws rather than being computed from microphysical condensation and settling processes coupled to atmospheric circulation. This simplification means that the retrieved cloud properties may not represent physically realizable particle distributions.
- **Chemical equilibrium assumptions:** Many models assume chemical equilibrium, which may be violated by photochemistry or vertical mixing. Carter et al. acknowledge that their chemical equilibrium assumptions could be violated by photochemistry, particularly for species like SO_2 that may be photochemically produced (Carter et al. 2024). Distinguishing equilibrium from disequilibrium scenarios remains challenging.
- **Cloud-gas degeneracies:** The strong coupling between cloud properties and molecular abundances means that inferred metallicities, C/O ratios, and elemental abundances can vary by orders of magnitude depending on cloud model choices (Lueber et al. 2024). Grey versus non-grey parameterizations produce drastically different H_2O abundances, and non-grey clouds can spuriously mimic water features.
- **Model-dependent molecular detections:** While H_2O and CO_2 are robustly detected across all instruments and models, other species show strong instrument and model dependence. For example:
  - NIRISS data require H_2O and K but do not detect CO_2 (Lueber et al. 2024)
  - NIRSpec G395H consistently detects H_2O and CO_2; SO_2 is only required with non-grey cloud models
  - NIRSpec PRISM detections of CO, SO_2, and H_2S depend critically on cloud assumptions; CH_4 is not detected

  This instrument-dependent detectability, combined with the SO_2 modeling debate, means that claimed detections beyond H_2O and CO_2 should be regarded as tentative.
- **Data-reduction:** As demonstrated by Schleich et al., posteriors for some molecules (especially SO_2 and C_2H_2) are sensitive to extraction pipeline choices, with posterior shapes becoming heavy-tailed or unconstrained depending on the input spectrum (Schleich et al. 2024). This suggests that current uncertainties may be underestimated for weakly detected species.

In summary, while JWST observations clearly reveal a cloudy, metal-rich atmosphere dominated by H_2O and CO_2, the detailed composition—particularly sulfur versus silicon chemistry—and precise abundance ratios remain uncertain due to cloud-gas degeneracies and model assumptions. These limitations motivate continued model development and careful assessment of systematic uncertainties as JWST data quality continues to improve.

4.3. Our Model Setup and Assumptions

For the retrievals we used TauREx-3 (Al-Refaie et al. 2021), a modular Bayesian framework combining forward atmospheric modeling with nested sampling. In the beginning we sued the course's Docker image. Due to computational limitations with nested sampling convergence times (~7 hours per retrieval), we trasnfered our analysis to a dedicated CloudVeneto virtual machine using the CA25_base image for an iterative refinement process. Key configurations:

Atmospheric Structure:

- 30 logarithmically spaced pressure layers (1–10^6 Pa)
- Hydrostatic equilibrium
- Isothermal temperature profile (single free parameter T)
- H_2–He dominated bulk composition ($He/H_2 = 0.25$, fixed)

Opacity Sources:

- Molecular cross-sections: ExoMol (H_2O, CO_2, CO, SO_2), HITRAN (trace species)
- Collision-induced absorption: H_2–H_2, H_2–He (HITRAN CIA database)
- Rayleigh scattering: H_2, He (wavelength^{-4} dependence)

Retrieved Parameters:

- Planetary radius R_p (uniform prior: 1.0–1.6 R_J)
- Isothermal temperature T (varied across retrieval iterations)
- Molecular volume mixing ratios: log-uniform priors spanning 10^{-12}–10^{-1}

Nested Sampling: The `nestle` algorithm with 100 live points computed posterior distributions via dynamic exploration of parameter space. This approach efficiently handles multi-modal posteriors and provides robust evidence estimates for model comparison.

Model Assumptions: For this course project, we accepted the following limitations; these were sufficient for the educational scope of the retrieval workflow, but for quantitive comparison with literature more sophisticated models would be required.

- **Isothermal Approximation:** Real atmospheres exhibit vertical temperature gradients driven by radiative heating/cooling and atmospheric circulation. An isothermal profile returns an effective average temperature weighted by opacity contributions, which does not correspond to any single physical layer. This simplification enables rapid computation but systematically biases retrieved temperatures and introduces degeneracies with molecular abundances.
- **Chemical Equilibrium:** We assume instantaneous chemical equilibrium at the retrieved temperature, neglecting vertical mixing (parametrized by K_{zz}) and photochemistry. WASP-39b's SO_2 detection indicates active photochemistry (Carter et al. 2024), which our model cannot capture.
- **Cloud-Free Atmosphere:** No cloud or haze opacity is included. Aerosols would reduce spectral feature amplitudes through wavelength-dependent scattering/absorption, cause degeneracy with temperature and molecular abundances. As mentioned in (Carter et al. 2024), there are indications that clouds are present in WASP-39b's observed spectrum as there are muted features. That means our retrieved parameters may be biased.
- **1D Geometry:** We assume azimuthal and latitudinal homogeneity. So thermal gradients or compositional were not captured, as onyl 1D models were used.

4.4. MASCARA-3b: Synthetic Spectrum

Input parameters:

- $R_p = 1.272\ R_J$, $M_p = 5.18\ M_J$
- $T = 1458$ K (isothermal)
- Chemistry: H_2O ($10^{-3.92}$ volume mixing ratio), CH_4 (10^{-5}), CO_2 ($10^{-3.2}$), CO ($10^{-3.1}$)
- Stellar parameters: $R_\star = 1.513\ R_\odot$, $T_{\rm eff} = 6347$ K

The retrieval attempted to recover R_p, T, and H_2O abundance while keeping other species fixed. Results:

- $R_p = 1.272 \pm 0.0004\ R_J$
- $T = 1456 \pm 70$ K
- $\log(H_2O) = -3.92 \pm 0.22$

The transmission and retrieval spectra are shown in Fig.[1] and in Fig.[2] the posterior distributions of the retrieved parameters.

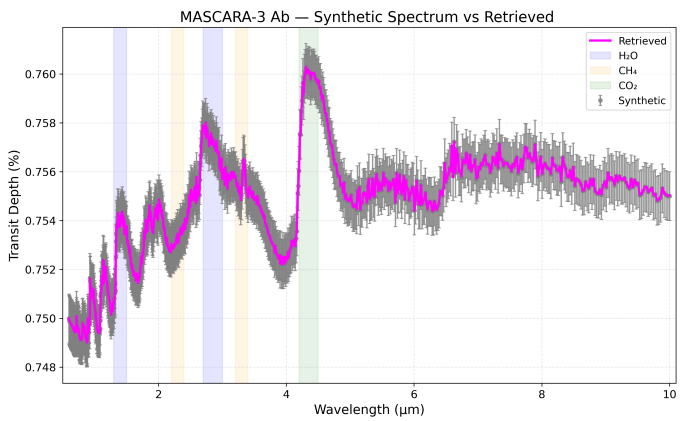

Fig. 1: Retrieved spectrum of MASCARA-3b.

As seen in the corner plot, the synthetic retrieval recovered input parameters with high precision and tight posterior distributions:

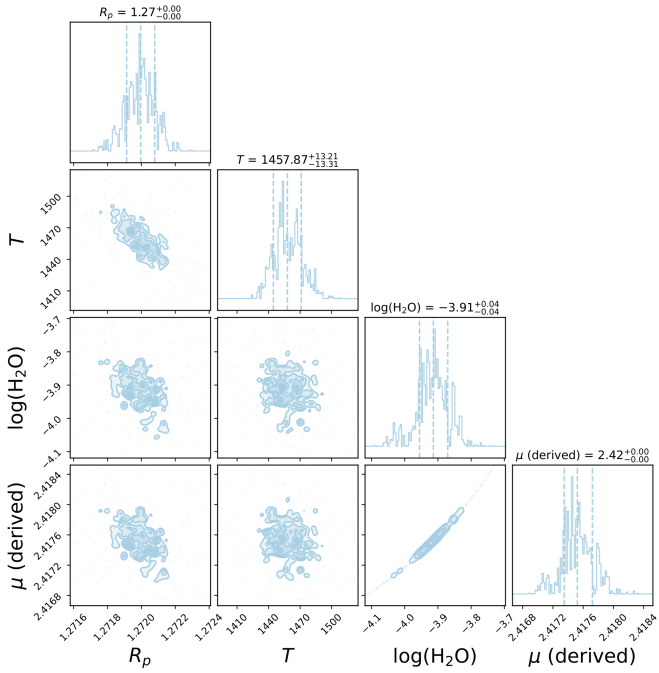

Fig. 2: Posterior distributions for retrieval of MASCARA-3b.

- **Planetary radius:** $R_p = 1.272 \pm 0.0004\ R_J$ (input: $1.272\ R_J$). The posterior is narrowly peaked with negligible asymmetry, confirming the radius is well-constrained by the transit depth.
- **Temperature:** $T = 1456 \pm 70$ K (input: 1458 K). The Gaussian-like posterior demonstrates excellent recovery of the input value within 1σ.
- **H_2O abundance:** $\log(H_2O) = -3.92 \pm 0.22$ (input: -3.92). Perfect recovery confirms correct opacity handling and radiative transfer implementation.

The on-point agreement confirms correct implementation of the forward model, opacity handling, and nested sampling. Posterior distributions show tight constraints with minimal parameter covariances, as expected for high signal-to-noise synthetic data. This successful recovery validates our retrieval setup before application to noisy observational data.

4.5. WASP-39b: Retrieval Process

Initial Attempt

Retrieving atmospheric parameters from real JWST spectra was more challenging than the synthetic validation case; we had to run multiple diagnostics in order to interpet the results correctly.

Initially, Na and K were included in the retrieval following literature reports of their presence in hot-Jupiter atmospheres. However, their strongest resonance lines lie in the optical, outside the wavelength range covered by the NIRSpec G395H data used here. As a result, their abundances could not be constrained, and the retrieval converged to a degenerate solution with enhanced Na/K and suppressed H_2O. Revisiting the reference study showed that the Na and K detections relied on additional optical observations, explaining why they cannot be robustly retrieved from the present dataset alone.

In our second attempt, the retrieved parameters converged at the prior boundaries; this indicated that the model was attempt-

ing to explore beyond the allowed parameter ranges but was constrained by the imposed priors. For confirmation, inspection of the spectrum did show that the model under-predicted the depth of the 4.4 μm CO_2 absorption band. More specifically:

- T = 900–2000 K (equilibrium temperature $T_{eq} \approx 1100$ K)
- H_2O: 10^{-12}–10^{-1} (0.001%–10%)
- CO_2: 10^{-12}–10^{-2} (0.001%–1%)
- CO, SO_2: 10^{-12}–10^{-2}

The retrieval converged with:

- T = 1588 K (**upper boundary**)
- $CO_2 = 2.3 \times 10^{-3}$ (0.23%, **upper boundary**)
- $H_2O = 8.1 \times 10^{-2}$ (8.1%)

So, convergence of both temperature and CO_2 abundance at their upper limits indicated need for change of the bounds. As confirmed from the literature, JWST studies (Carter et al., 2024) show that WASP-39b has a high atmospheric metallicity, about 10× solar. In a solar-composition atmosphere, CO_2 is around 0.04%, so 10× metallicity gives 0.4%, which is higher than the 1% maximum we had used.

Final results

After adjusting the bounds more generously, here are the updated priors accounting for high metallicity:

- T = 800–1300 K (allowing cooler terminator regions)
- H_2O: 10^{-5}–2×10^{-1} (20%)
- CO_2: 10^{-5}–10^{-1} (10%)
- CO, SO_2: 10^{-12}–10^{-2} (unchanged)

Retrieved:

- $T = 805 \pm 104$ K (**lower boundary**)
- $H_2O = (6.3 \pm 1.2) \times 10^{-2}$ (6.3%)
- $CO_2 = (1.4 \pm 0.35) \times 10^{-3}$ (0.14%, no longer at boundary)
- CO = $< 2 \times 10^{-4}$ (poorly constrained)
- $SO_2 = < 6 \times 10^{-4}$ (poorly constrained)

CO_2 successfully moved away from the boundary, improving the spectral fit at 4.4 μm. However, temperature moved to the *lower* boundary, revealing a new issue. Also, looking at the spectrum revealed that at 2.8–2.9 μm the model under-predicts H_2O absorption depth and at 4.4 μm it still slightly under-predicts CO_2 absorption, though improved from initial attempt.

In figures Fig.[3] and Fig.[4] we can see the retreived spectrum and the corner plots for the retrieval of WASP-39b.

Well-Constrained Parameters:

- **Planetary radius:** $R_p = 1.22 \pm 0.01\ R_J$. The posterior has mainly one peak and is well-separated from prior boundaries, indicating robust constraint from the observed transit depth.
- **H_2O abundance:** $\log(H_2O) = -1.20 \pm 0.13$ (5.0–6.3%). The posterior peaks away from boundaries with asymmetric uncertainties reflecting the log-uniform prior structure. Strong constraints arise from multiple H_2O absorption features at 2.7–3.0 μm.
- **CO_2 abundance:** $\log(CO_2) = -2.86 \pm 0.30$ (0.11–0.14%). Despite initial boundary issues, the expanded prior allowed the posterior to peak at physically reasonable values consistent with high atmospheric metallicity.

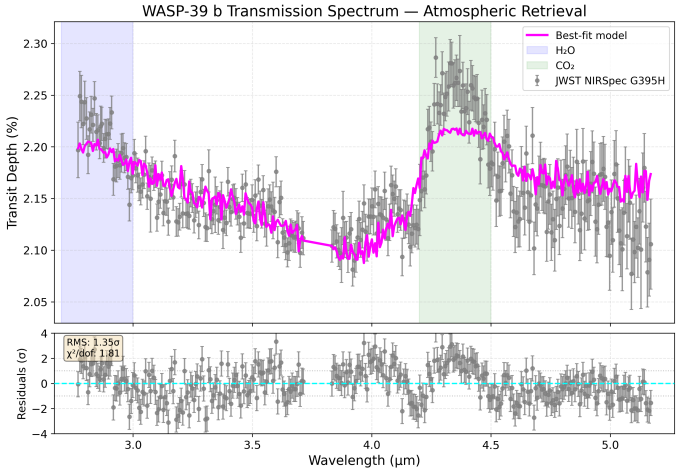

Fig. 3: WASP-39b fitted spectra.

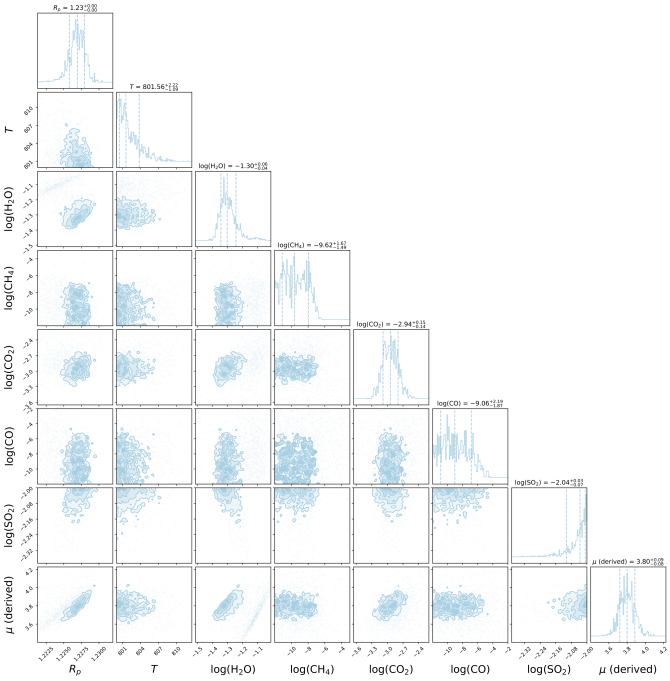

Fig. 4: Posterior distributions for retrieval of WASP-39b.

Poorly Constrained Parameters:

- **Temperature:** $T = 805 \pm 104$ K. The posterior shows a sharp cutoff at the lower prior boundary (800 K), indicating that the model seeks cooler temperatures but is constrained. The large positive uncertainty (+104 K) versus tight negative uncertainty (−4 K) shows asymmetry because of the boundaries. This suggests the isothermal assumption is inadequate for capturing the true atmospheric thermal structure.
- **CH_4, CO, SO_2:** Posteriors for trace species are extremely broad (2–5 orders of magnitude) and show negligible information gain from the data. This indicates weak or absent spectral features in the NIRSpec G395H range, providing no meaningful constraint on these abundances. Retrieved values represent upper limits rather than detections.

Parameter Correlations:

- **Temperature-H_2O anti-correlation:** Lower temperatures correlate with higher H_2O abundances. Both affect atmo-

spheric scale height $H \propto T/\mu g$ and thus spectral feature amplitudes. This degeneracy cannot be broken without additional constraints (e.g., non-isothermal $T(P)$ profiles or cloud opacity that independently affects continuum levels).

- **Temperature-CO_2 correlation:** Warmer temperatures slightly favor higher CO_2 abundances to fit the 4.4 μm feature depth. However, this correlation is weaker than the T-H_2O degeneracy due to CO_2's more localized spectral signature.
- **H_2O-CO_2 independence:** Minimal correlation between water and carbon dioxide abundances indicates their spectral features occupy distinct wavelength regions (2.7–3.0 μm vs. 4.4 μm), allowing independent constraints.

The Nestle sampler achieved convergence with 2615 total samples and 2217 valid samples (85% efficiency). The evidence estimate stabilized after ~1500 samples, confirming adequate posterior exploration. All parameters show smooth, unimodal posteriors (except temperature hitting the boundary), validating numerical convergence.

In Fig.[5] we can also see the opacity contributions from different atmospheric components in WASP-39b's retrieved model. CO_2 (green) and H_2O (red) dominate absorption at 4.3–4.5 μm and 2.7–5.2 μm respectively, while CH_4 and CO contribute negligibly, explaining their poor retrieval constraints.

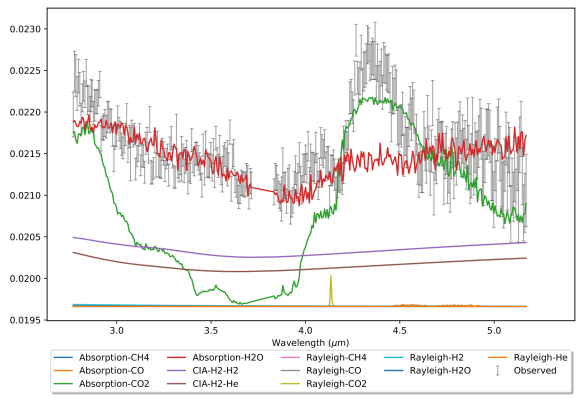

Fig. 5: Molecular contributions in WASP-39b.

Literature Comparison

From the posterior distributions of WASP-39b we can make the following observations, comparing it to the documentation we based this report on, by Carter et. al:

Parameter	This project	Carter_2024	vs
R_p (R_J)	1.22 ± 0.01	1.27 ± 0.04	Within 2σ
T (K)	805 ± 104	~1100	Offset
H_2O (%)	6.3 ± 1.2	Few %	Consistent
CO_2 (%)	0.14 ± 0.35	Detected	Consistent
CO (%)	< 0.02	Detected	Marginal
SO_2 (%)	< 0.06	Detected	Marginal

Table 5: Retrieved vs Carter et al. (2024)

The retrieved isothermal temperature (~805 K) is significantly lower than both the equilibrium temperature (~1120 K) and the ~1100 K reported by Carter et al. (2024). This offset arises for the following reasons: The isothermal model we used returns a pressure-weighted average temperature rather than a true vertical $T(P)$ structure. Also, because of the well-known degeneracy between temperature, molecular abundances, and clouds: cooler temperatures combined with higher abundances can produce similar spectral feature amplitudes to warmer, lower-abundance solutions through their effect on the scale height $H \propto T/\mu g$. In addition, the absence of cloud opacity in our model forces the retrieval to suppress spectral features by lowering the temperature, whereas Carter et al. (2024) showed that including clouds raises the inferred temperature by ~200 K. Limited wavelength coverage further prevents robust constraints on trace species and alkali metals that were detected in the multi-instrument JWST analysis of Carter et al. (2024). Residual mismatches in the H_2O and CO_2 bands also indicate the inadequacy of a single-temperature atmosphere, as these molecules absorb at different pressure levels and therefore sample layers with different thermal conditions; an effect that is reduced when using the non-isothermal retrieval approach implemented by Carter et al. (2024).

So, these biases are documented in the retrieval literature and cannot be resolved without more complex atmospheric models.

The big contrast between MASCARA-3b's tight, uncorrelated posteriors and WASP-39b's broad, degenerate distributions shows that with real data some problems arise:

- Observational noise broadens uncertainties by 1–2 orders of magnitude
- Systematic effects (stellar contamination, detector systematics) introduce correlations absent in synthetic data
- Model limitations (isothermal profile, missing clouds) manifest as boundary-hitting and residual degeneracies
- Limited wavelength coverage reduces information content, so trace species don't have properly updated bounds.

This demonstrates that synthetic validation, cannot predict the full complexity of real atmospheric retrievals, even though it can be used to confirm the code's robustness. Also, refining parameters in iterations based on posterior diagnostics (boundary-checking, correlation analysis, residual inspection) is necessary to extract physically meaningful constraints from JWST observations.

5. Conclusions

This project integrated machine learning transit detection with Bayesian atmospheric retrieval to explore both the identification and characterization of exoplanets. The workflow included automated classification of Kepler light curves and iterative atmospheric inference from JWST transmission spectra.

5.1. Machine Learning Performance and Limitations

SVMs and Neural Networks reached 99% precision on clean Kepler data, showing that Fourier preprocessing captures strong periodic transit signals effectively, but performance dropped to ~56% for weak injected transits, showing clear limitations. Fourier features preserve periodicity while discarding transit-shape information, suggesting that extracting physical transit parameters would improve sensitivity to shallow signals. Severe class imbalance and limited training data further affect performance, with Neural Networks prone to overfitting while SVMs

remain more robust in small-sample regimes. The resulting precision on weak transits shows that there is certain difficulty for real exoplanet detection, which is consistent with false-positive rates that are reported in professional vetting pipelines.

5.2. Atmospheric Retrieval

TauREx-3 validation on MASCARA-3b synthetic spectra confirmed correct implementation, recovering injected parameters (radius, temperature, and H_2O abundance) with high precision. Applying the framework to real WASP-39b JWST data required iterative refinement, as initial priors were too restrictive for its high-metallicity atmosphere, causing temperature and CO_2 to converge at prior boundaries. Expanding priors based on literature metallicity constraints (~10× solar) released these parameters and improved spectral fits. The retrieved temperature of 805K is about 300K cooler than expected from literature ((~1100K) because the isothermal, cloud-free model cannot account for vertical temperature variations, day–night terminator differences, or cloud effects, resulting in well-known degeneracies between temperature and molecular abundances. Despite these simplifications, H_2O (6.3%) and CO_2 (0.14%) were robustly retrieved in agreement with Carter et al. (2024), while trace species such as CO and SO_2 remained unconstrained because of limited wavelength coverage.

5.3. Insights into WASP-39b's Atmosphere

Our retrieval successfully confirmed WASP-39b's water-rich (H_2O: 5.0–6.3%) and metal-enhanced (~10× solar metallicity) atmosphere, with retrieved planetary radius (1.22–1.23 R_J) and CO_2 abundance (0.11–0.14%) consistent with literature values (Carter et al. 2024). The combination of low mass (0.28 M_J), high metallicity, and extremely inflated structure (density ~0.14 g cm^{-3}, scale height ~1600 km) supports formation by core accretion at large orbital distances (>5 AU) where water ice is abundant, followed by inward migration to its close-in orbit (0.0486 AU). When combined with broader JWST results (including subsolar C/O ratio (0.35) indicating oxygen enrichment and detection of photochemically-produced SO_2 (Carter et al. 2024)), WASP-39b serves as a benchmark for understanding giant planet formation, migration, and active atmospheric chemistry driven by stellar UV radiation. However, our single-instrument analysis (NIRSpec G395H: 2.77–5.17 μm) could not constrain trace species or alkali metals, demonstrating that more accurate atmospheric characterization requires observations that extend across optical to mid-infrared wavelengths. The systematic temperature offset (805 K vs. 1100 K literature) shows that simplified isothermal, cloud-free models introduce biases that more complex $T(P)$ profiles and cloud parameterizations can reduce.

5.4. Improvements Suggestions

There are be many ways in which both the transit detection methodology and atmospheric characterization presented in this work can be improved. On the machine learning side, extracting physical transit parameters via model fitting rather than relying solely on Fourier features would provide more interpretable inputs, while implementing ensemble methods combining SVM and neural network predictions could improve robustness. Exploring one-dimensional convolutional architectures designed for time-series analysis and applying these methods to larger datasets with confirmed false positives (e.g., TESS) would enable more rigorous validation.

For atmospheric retrieval, the analysis could be enhanced by utilizing the full JWST wavelength coverage (0.5–5.2 μm) rather than single-instrument subsets, implementing non-isothermal temperature-pressure profiles, and including cloud or haze opacity. Carter et al. (2024) demonstrated that incorporating cloud opacity shifts retrieved temperatures upward by approximately 200 K for WASP-39b, highlighting the importance of aerosol modeling. Another suggestion would be to apply a Bayesian retrieval framework using machine learning, where radiative transfer codes like petitRADTRANS (Mollière et al. 2019) precompute synthetic spectra on parameter grids, and trained neural networks enable efficient nested sampling for robust uncertainty quantification and model comparison (Lueber et al. 2024).

5.5. Future missions

The ongoing space missions as well as the upcoming ones are showing the importance of combining machine-learning detection with atmospheric retrieval techniques. **PLATO** (launch 2026) will discover thousands of transiting planets around bright stars, making automated ML vetting pipelines essential for prioritizing the most promising targets for atmospheric follow-up. **Ariel** (launch 2029) will perform population-level atmospheric surveys of ~1000 exoplanets, where tools like TauREx will enable consistent analysis across many systems and reveal trends in atmospheric composition, clouds, and how planets form. **JWST** continues to characterize individual benchmark targets; understanding retrieval degeneracies and model limitations, as demonstrated in the WASP-39b analysis, is crucial for interpreting transmission spectra.

In the upcoming missions, data quaility and size will grow and so integrating machine learning methods will help address fundamental questions about exoplanetary research and it will become increasingly important.

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